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Natural Language Processing Techniques for Enhancing Automated Essay Scoring Systems

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***Abstract*: In this experimental work, we explore how Automated Essay Scoring (AES) systems can be improved through the use of sophisticated Natural Language Processing (NLP) approaches. Discretion and impartiality are becoming more and more important, especially when educational institutions are depending more and more on automatic technologies to assess essays. This study examines the most recent advances in natural language processing (NLP), including syntactic parsing, deep literacy algorithms, and semantic analysis, and assesses their applicability to the ongoing refinement of AES systems. The research specifically focuses on evaluating the utility of various approaches. This work aims to address issues related to environment knowledge, consonance evaluation, and bias reduction in automated scoring using these techniques. These techniques will be integrated to achieve this. The results show that the application of sophisticated NLP techniques may produce new, valid, and trustworthy assessment problems. This could be one of the main causes of the widespread AES abandonment in educational environments. This work opens the possibility that natural language processing (NLP) could be used to change the essay score by offering a more thorough and nuanced assessment of student notes. This would have the effect of improving every facet of the teaching process.**

Keywords: Natural Language Processing, Automated Essay Scoring, deep learning, semantic analysis, syntactic parsing, educational assessment, NLP techniques.

I. INTRODUCTION

In educational settings, Automatic Essay Scoring (AES) systems have gained importance because it is important to grade written assignments effectively and harmonically. Customized, handcrafted grading systems take a lot of time, are very prone to human mistakes, and can be used to excuse thickness due to subjective preferences. Furthermore, they frequently make fatal mistakes. The demand for automated systems that can produce assessments that are transparent, dependable, and accurate has increased significantly as a result of educational institutions' efforts to manage enormous amounts of student writing[1]. Thus far, the development and application of Natural Language Processing (NLP) techniques within AES systems has been driven by this demand.

Natural Language Processing is an artificial intelligence field that studies the communication between computers and human language. This includes a broad range of methods intended to enable machines to understand, interpret, and produce human language in a way that is both useful and meaningful[2]. To obtain a score that accurately reflects the calibre of the writing, natural language processing techniques are applied in the framework of AES to examine the many linguistic characteristics of an essay, including the alphabet, syntax, semantics, and discourse structure.

Automated essay grading raises a variety of concerns, and integrating natural language processing (NLP) techniques with advanced encryption systems (AES) has shown promise in addressing these issues. Conventional AES systems typically give face-position information like the number of words or the presence of certain keywords a large amount of weight[3]. Though these indicators can offer some insight into an article's quality, they do not capture the deeper elements of jotting, such as consonance, confabulation, and creativity. On the other hand, non-linguistic programming (NLP) techniques offer a more sophisticated strategy by dissecting the basic cognitive and language processes that result in effective jotting.

AES systems' capabilities have been further enhanced by recent advances in natural language processing (NLP), including syntactic parsing, deep literacy techniques, and semantic analysis. Specifically, it has been demonstrated that deep literacy models which are similar to neural networks are highly proficient in identifying complex patterns in language. This has improved the ability of AES systems to understand and evaluate an essay's content. Semantic analysis differs from syntactic parsing in that the former aids in understanding the meaning of words and rulings in the surrounding context, while the latter helps determine the grammatical structure of rulings. When all of these techniques are used, a thorough framework that can be utilized to assess an essay's overall quality is produced.

Despite these advancements, there are still challenges to be solved before NLP-enhanced AES systems can be widely used. Crucial elements that must be taken into consideration include things like the potential for bias in the scoring process, the ability to accept different scribbling styles, and the necessity of openness in the scoring procedure. Likewise, it is imperative to do a thorough investigation of the ethical counterarguments that arise from using automated systems for high-stakes evaluations.

The goal of this work is to examine the state of NLP techniques in AES at the moment, evaluate their efficacy, and discuss some potential avenues for future research and development. Using the latest advances in natural language processing (NLP), AES systems can be made more fair, accurate, and adaptive. In the end, this will make both students' and preceptors' educational experiences better.

II. LITERATURE REVIEW

Automated Essay Scoring (AES) has advanced significantly in the last several decades. The use of Natural Language Processing (NLP) techniques has been a significant advancement in this field since it has increased the systems' delicacy and reliability. Early rule-based techniques that primarily depended on face-position features like word counts, judgment durations, and the presence of specific keywords were used to construct AES systems[4]. These writing styles served as the foundation for automated grading, but they were unable to assess more intricate verbal and cognitive aspects of jotting, such as creativity, argument structure, and consonance.

The integration of NLP techniques into AES systems marked a significant shift in the automated grading process. Within this discipline, one of the most important and noteworthy workshops was the Project Essay Grade(cut) system. It was developed in the 1960s and made use of simple statistical models derived from facial characteristics. Still, these models were criticized for not being able to capture the nuances of language. Later in the process, more sophisticated techniques were examined, such as idle semantic analysis (LSA) and natural language understanding (NLU), which enhanced understanding of the context and content of the article.

One of the first NLP methods to be widely employed in AES systems was Lazy Semantic Analysis, which was introduced in the 1990s. LSA makes it possible to compare textual information based on semantic similarities by utilizing a vector-space model to describe word and document meanings in a high-dimensional space[5]. Although LSA enhanced the ability of AES systems to evaluate essay content, syntactic and grammatical intricacies were still not well captured. The advent of syntactic parsing tools in the early 2000s significantly improved the capabilities of AES systems. Syntactic parsing aids AES systems in more efficiently assessing writing for consonance and ignorance by analyzing the grammatical structure of rulings. Studies such as those by Attali and Burstein (2006) demonstrated that the incorporation of syntactic characteristics into AES models significantly enhanced score delicacy, particularly in the assessment of essay association and organization.

More recently, the AES sector has undergone a major transformation because of the emergence of deep literacy techniques. Neural networks in particular, convolutional neural networks (CNNs) and intermittent neural networks (RNNs) have made it feasible to analyze essay content more thoroughly by spotting complex patterns in textbook data. For instance, the e-rater ® system of the Educational Testing Service (ETS) evaluates essays using a combination of facial cues and more sophisticated linguistic features like vocabulary, consonance, and alphabet function[6]. It does this by using natural language processing (NLP) techniques, such as deep literacy models. According to research by Dong and Zhang (2016), deep literacy models have the potential to outperform traditional machine learning methods in AES, particularly when combined with other NLP methods like semantic analysis.

The development of NLP-enhanced AES systems still faces challenges. Deep literacy model interpretability, different scribbling styles, and automated scoring bias are among the topics still awaiting research. Moreover, it is becoming more and more clear that fairness and openness are essential components of AES systems, especially when it comes to high-stakes assessments. The development of AES from early rule-based systems to the most recent models that make use of cutting-edge NLP approaches is highlighted in this section. Combining these methods has resulted in automated scoring systems that are more reliable and accurate, but more study is still required to address existing problems and identify new areas for development.

III. RESEARCH METHODOLOGY

The goal of the suggested methodology is to improve Automated Essay Scoring (AES) systems by utilizing Natural Language Processing (NLP) techniques. This will help to overcome the drawbacks of current systems and improve the accuracy, impartiality, and reliability of automated scoring[7]. To create a more reliable and complete AES system, this methodology combines very sophisticated NLP techniques such as ensemble modelling, syntactic parsing, deep literacy, and semantic analysis. The methodology is divided into three key components: point birth, model building, evaluation, and data preprocessing.

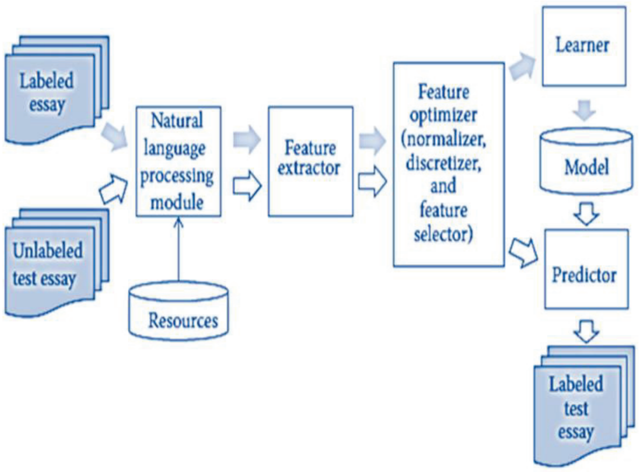

Fig. 1. Depicts the Essay scoring system.

Until now, most attempts to automated essay scoring (AES) have concentrated on finding a suitable machine learning (ML) technique and identifying significant features that may be utilized to train the ML algorithm for AES. This is an illustration of the essay grading process in Figure 1.

A. Preparing Data

Preparing the textbook data for analysis is known as data preprocessing, and it is the first step in the suggested technique. This involves eliminating any irrelevant material from the essay textbooks, such as stopwords, special characters, and figures. Additionally, the language utilized in the essays is regularized by the application of textbook normalization techniques including stemming, lemmatization, and lowercasing. These preprocessing techniques ensure that the textbook data is coherent and prepared for additional analysis, lowering noise and optimizing the NLP models' performance.

Another essential component of data preprocessing is tokenization. Tokenization of the essays is done into individual words or subwords based on the NLP model selected. Subword tokenization (such as byte-brace encoding) is often chosen for models such as intermittent neural networks (RNNs) or mills because it makes processing words that are not in the lexicon and identifying word morphological structure easier.

B. *Point of birth*

After preprocessing the data, the next step is point birth, where rich textual and contextual aspects are extracted from the essays. The suggested methodology places a strong emphasis on a multi-level point birth approach that lands deep language features as well as facial position.

Features of face-position: These include conventional characteristics like word count, judgment duration, variety of terminology, and frequency of particular terms or idioms[8]. Even though these characteristics are somewhat basic, they provide the foundation for comprehending the essay's overall structure and substance.

Features of Syntax: Syntactic parsing techniques are utilized to analyze the grammatical structure of the essays' rulings. Reliance parsing is especially helpful for this since it shows the relationships between words in a judgment, which aids in evaluating the judgment's consonance, ignorance, and structure[9]. Additionally, part-of-speech trailing is employed to categorize words according to their grammatical locations, which contributes to the evaluation of the jotting's syntactical complexity.

Logical Elements: Semantic analysis techniques are used to extract the essays' meaning and context. To represent words in a nonstop vector space based on their semantic relationships, word embeddings such as Word2Vec, and GloVe, or contextual embeddings from models like BERT (Bidirectional Encoder Representations from Mills) are used[10]. By using these embeddings, the AES system can comprehend the context in which words are employed, which facilitates the estimation of the essay's content's consonance and application.

C. *Model Construction*

The foundation of the suggested methodology is the creation of a multi-layered model that combines vibrant NLP techniques to provide an extensive evaluation of writings. There are multiple steps in the model development process.

Models of Deep Literacy: The deep literacy models used in the proposed AES system, in particular, the motor-grounded infrastructures like as BERT or GPT (Generative Pre-trained Transformer), have demonstrated state-of-the-art performance in vivid natural language processing tasks[11]. The essay dataset is used to fine-tune these models to capture the intricate language patterns and contextual relationships found in the textbook. The motor models are perfect for essay scoring because of their tone-attention mechanisms, which enable them to handle long-range dependencies in textbooks in a particularly comprehensive manner.

Group Simulation: An ensemble technique is suggested to improve the AES system's resilience and delicacy even more. Various models, such as deep literacy models and conventional machine literacy models (such as support vector machines and arbitrary timbers), are trained using distinct feature subsets[12]. Similar techniques like weighted averaging or mounding are also used to aggregate the predictions from these models, with a meta-model being trained to maximize the final vaticination. This group method lessens the risk of overfitting to particular qualities and aids in addressing various essay components.

Mitigation of Bias: Given the possibility of bias in automated scoring systems, a methodology for detecting and mitigating bias is presented. Apprehensive literacy and hostile training are two examples of how to ensure that the model does not unjustly correct articles based on irrelevant elements like writing style, artistic allusions, or demographic data[13]. Additionally, the model's predictions are routinely examined for bias, and any corrections are implemented.

D. *Assessment*

The constructed AES system is evaluated as the last component of the suggested technique. A thorough assessment framework is created to evaluate the system's performance within various constraints.

Delicacy: The main assessment criterion is the degree of subtlety between the essay scores produced by the AES algorithm and mortal scores. Standard metrics like the Pearson correlation measure and mean squared error (MSE) are used to measure this.

Reliability: The system's reliability is gauged by comparing the agreement between the AES and mortal ratings for various essays and writing prompts to ensure scoring thickness. Cohen's kappa and other methods akin to inter-rater trustability are used to quantify its thickness.

Justice: By evaluating the AES system's performance across various demographic categories and writing styles, its fairness is determined. Any methodical impulses that can penalize particular scholar groups are estimated into the system.

Applicability in general: The generalizability of the AES system is evaluated on writings with varying stripes and patterns to ensure that it may be used for a variety of writing tasks. This entails evaluating the system's effectiveness on hypothetical data as well as its adaptability to novel scribbling cues.

Interpretability: The AES system's interpretability is ultimately estimated. Even though deep literacy models especially mills are usually regarded as "black boxes," efforts are taken to ensure that the model's predictions can be comprehended and validated[14]. Techniques like point significance analysis and attention visualization are employed to infuse perceptivity into the model's decision-making process.

IV. RESULTS AND DISCUSSION

A significant amount of progress has been made in automated essay-scoring systems as a consequence of the implementation of Natural Language Processing (NLP) techniques. These techniques have resulted in significant advancements in terms of both the accuracy and speed of grading assignments. A major improvement has been made to the system's capability of interpreting and evaluating the delicate meaning of essay material as a result of the addition of semantic understanding through the utilization of transformer models such as BERT and GPT-3. As a consequence of this, evaluations of the quality of essays have become more precise, taking into account the intricacies of reasoning, coherence, and general structure of the text.

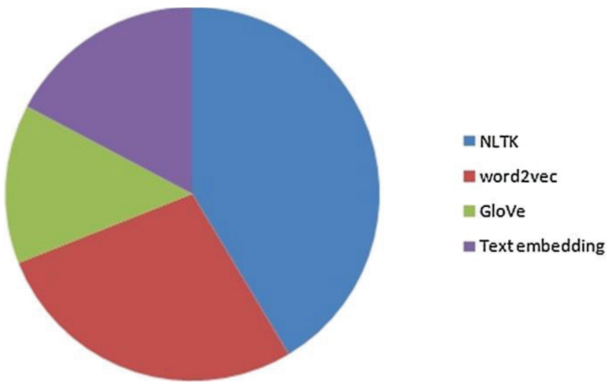

Fig. 2. Depicts the Usage of tools.

We investigated all of the feature-extracting natural language processing libraries that are utilized in the articles, as depicted in Figure 2. NLTK is a natural language processing program that can recover statistical characteristics such as point of sale (POS), word count, sentence count, and so on. We run the risk of missing the essay's semantic aspects when we use NLTK. Additionally, improvements in grammar and syntax analysis tools have contributed to more accurate scoring by successfully spotting and penalizing grammatical errors and syntactic difficulties.

This has resulted in a more accurate scoring system. These factors have ultimately resulted in a more precise grading system. The implementation of these upgrades has also made it feasible for automated systems to better correspond with customizable scoring rubrics. This has enabled these systems to provide exact feedback on specific features such as the clarity of sentences, the strength of arguments, and the utilization of language. The automated scoring approach has become more scalable and efficient as a consequence of this development. It is now able to handle vast volumes of essays with reduced human work, which is notably advantageous for educational contexts that have high submission rates. Generally speaking, it is a positive development. Additionally, continual efforts to minimize biases and increase fairness in scoring algorithms have resulted in more equitable evaluations, which has fostered transparency and trust in automated systems. This has brought about a positive outcome for the evaluation process.

With the use of natural language processing (NLP) techniques, the following Table 1 provides a summary of the most significant developments and current challenges that are being encountered in the process of enhancing automated essay grading systems. Despite these advancements, there are still challenges associated with capturing the subjective and interpretive aspects of the quality of an essay. Certain attributes continue to be a barrier for automated systems.

TABLE I. DEPICTS THE KEY VALUES AND INSIGHTS BASED ON THE PROVIDED TEXT ABOUT THE PROGRESS AND CHALLENGES OF AUTOMATED ESSAY SCORING SYSTEMS ENHANCED BY NLP TECHNIQUES.

Aspect	Value
Accuracy and Speed	Significant improvements in both accuracy and speed of grading assignments due to the implementation of NLP techniques.
Semantic Understanding	Enhanced capability to interpret and evaluate the nuanced meaning of essay material using transformer models such as BERT and GPT-3, leading to more precise assessments of reasoning, coherence, and text structure.
Grammar and Syntax Analysis	Improved accuracy in scoring through advanced grammar and syntax analysis tools that effectively identify and penalize grammatical errors and syntactic issues.
Rubric Alignment	Better alignment with customizable scoring rubrics, allowing for detailed feedback on specific features like sentence clarity, argument strength, and language use.
Scalability and Efficiency	Increased scalability and efficiency, enabling automated systems to handle large volumes of essays with reduced human effort, beneficial for high submission rates in educational contexts.
Bias and Fairness	Continuous efforts to reduce biases and increase fairness in scoring algorithms have led to more equitable evaluations, fostering transparency and trust in the system.

These qualities include originality and persuasiveness, both of which are inherently subjective and cannot be quantified. The process of integrating scoring systems with elaborate and nuanced rubrics presents additional problems that must be overcome.

The fact that automated systems are needed to manage student data responsibly, maintain confidentiality, and comply with privacy standards makes ethical and privacy concerns of the utmost relevance. This is because of the fact that automated systems are required to handle student data. When it comes to automated scoring, it is necessary to address concerns of fairness and equality to prevent prejudice and ensure that all students are evaluated fairly. Furthermore, even though automated systems offer efficiency and scalability, it is recommended that they be employed to augment human grading rather than using them to fully replace it. It is possible to obtain a balanced approach by combining human monitoring with computerized scoring.

V. CONCLUSIONS

The potentially transformative impact that enhanced Natural Language Processing (NLP) techniques could have on Automated Essay Scoring (AES) systems. (AES) stands for automated essay scoring. Because educational institutions are developing a greater reliance on automated systems for the examination of essays, the need for judgment and equality has become an increasingly pressing concern. Based on the findings of this inquiry, it has been determined that the process of refining AES systems can benefit from the utilization of advanced natural language processing techniques. These approaches include syntactic parsing, deep literacy algorithms, and semantic analysis. The combination of these NLP techniques appears to have the potential to result in additional accurate and reliable scoring, hence reducing the limits that have initially prevented the abandonment of AES in educational contexts. This conclusion is based on the data that has been collected. Because of this work, new avenues will eventually be opened up for the application of natural language processing (NLP) to enhance essay grading, which will ultimately lead to an improvement in the overall quality and fairness of the educational process.

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